

7 The Galapagos Islands are located off the west coast of South America.

The biodiversity on these islands has changed over the past 40 years due to human activity.

One of these islands is called Santa Cruz.

- (a) The population of Santa Cruz increases by approximately 6.4% per year.

The majority of the population, 85%, live in urban areas.

The population of Santa Cruz living in urban areas in 2020 was 17 000.

Predict the total population of Santa Cruz in 2025.

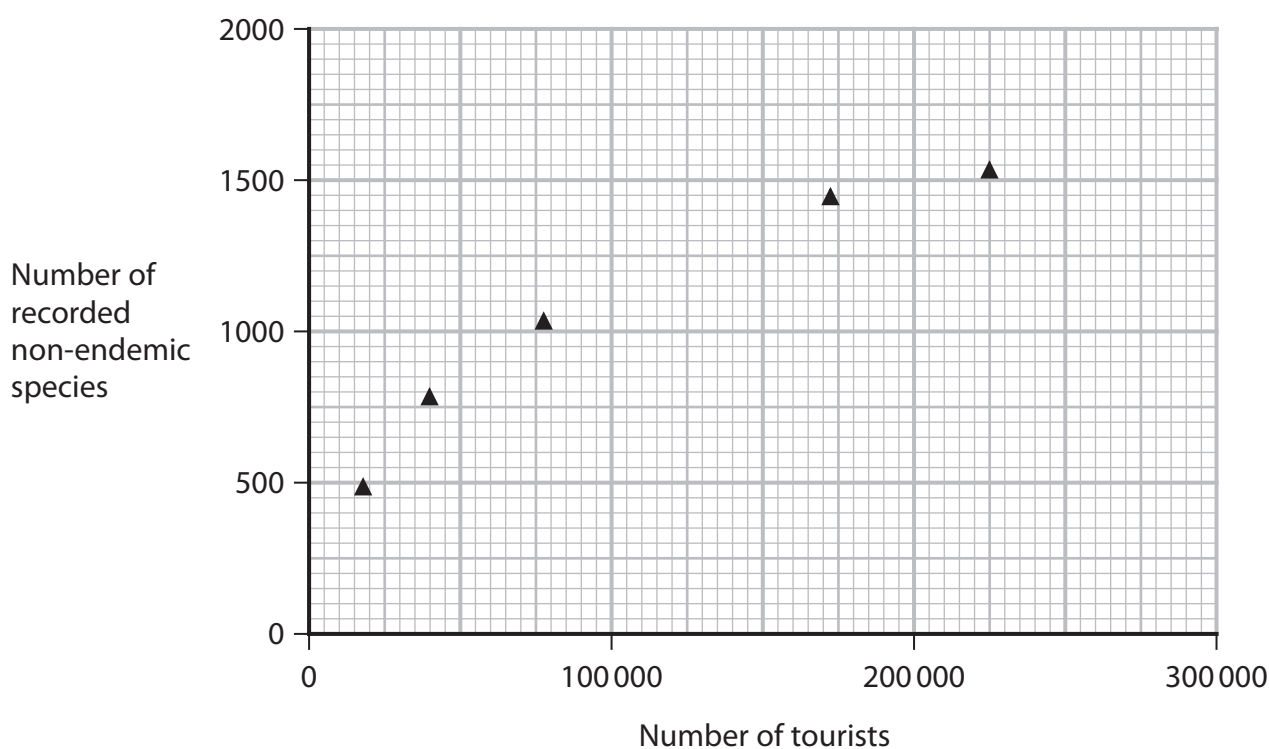
Assume that the rate of increase stays constant.

(3)

Answer

- (b) Non-endemic species have been introduced to the Galapagos Islands.

The graph shows the number of tourists to all the islands and the number of recorded non-endemic species.



<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0184379>



State the relationship shown in the graph.

(1)

- (c) The wild blackberry was introduced to Santa Cruz in 1970 and has since spread over much of the island.

This has resulted in the reduction of the native endemic forest.

In 1960, the number of species in an area of the native forest was measured.

The table shows these results.

Species	Number of individuals (n)	(n – 1)	n(n – 1)
A	21	20	
B	2	1	2
C	4	3	12
D	13	12	
E	54	53	
F	15	14	210
G	6	5	30
H	32	31	
	Total (N) =		$\Sigma n(n - 1) =$

- (i) Calculate the index of diversity (D) for this area of the forest using the formula

$$D = \frac{N(N - 1)}{\Sigma n(n - 1)}$$

Use the table to help you.

Give your answer to **one** decimal place.

(3)

Answer



- (ii) Explain why the spread of the wild blackberry affects the biodiversity of the island.

(5)

DO NOT WRITE IN THIS AREA

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(Total for Question 7 = 12 marks)

